

Multiple Choice (12 marks)

1. What is the main function of a plant stem?
 - (a) to collect pollen from other plants to make seeds
 - (b) to absorb water and plant nutrients from the soil
 - (c) to carry food and water to the rest of the plant
 - (d) to make food for the plant
2. How does refraction affect the look of things
 - (a) They look black
 - (b) They appear same
 - (c) They just disappear
 - (d) They appear different
3. The distance from Earth to the Sun is about 150,000,000 km. What abbreviation do scientists use to describe this distance?
 - (a) meters per second (m/s)
 - (b) light-year (ly)
 - (c) astronomical unit (AU)
 - (d) electromagnetic unit (EMU)
4. Which of the examples below has not been developed by the European Court of Human Rights as a situation where the state has a positive obligation to protect?
 - (a) When the applicant is in the custody of the state
 - (b) Where the state must investigate claims against it
 - (c) The development of a legal framework to protect those within the state
 - (d) To provide housing for all those homeless within a state
5. A disorder is associated with certain flaws in the human respiratory system. Which is most likely a typical symptom of this disorder?
 - (a) poor vision
 - (b) difficulty breathing
 - (c) reduced muscle mass
 - (d) increased saliva production

6. How long might a bear likely to remain its den without eating, drinking, or excreting after November?
- (a) The first few weeks of December
 - (b) Until it hears the call of the wild
 - (c) For around twenty weeks
 - (d) until hunters kill it for its pelt

True / False (12 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Transpiration involves' is 'the production of new oxygen molecules'.
2. True or False: The correct answer to 'soil is formed by' is 'the fumes from car exhaust'.
3. True or False: The correct answer to 'One of the first models of the physiology of the human sexual response was provided by' is 'William Masters and Virginia Johnson'.
4. True or False: The correct answer to 'Which of the following would push water out of its body to evade an orca?' is 'a squid'.
5. True or False: The correct answer to 'To determine the boiling point of a liquid, a student must use all of these tools except' is 'a stopwatch'.
6. True or False: The correct answer to 'Skin can protect your' is 'feelings'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to check whether it follows the sequence given in the patterns array.
2. Write a python function to calculate the sum of the numbers in a list between the indices of a specified range.

End of Paper